Assignment - 11

Instructions: Follow the same pattern as before —
1. Dry run each input on paper,
2. Write and test your code,
3. Submit both your dry run and final working code.

1. Stack With Undo Feature

Problem: Design a stack that supports the usual operations (push, pop, top) and an additional operation `undo(k)` which undoes the last `k` operations.

Input: Sequence of operations. **Output:** Final stack content.

Constraints:

- 1 <= operations <= 10^5
- Only last `k` operations are undone in reverse order.

Example: Input: push(5), push(2), pop(), push(7), undo(2) Output: [5, 2]

2. Bracket Validation with Priority Levels

Problem: Given an expression containing brackets of different types, validate it ensuring higher-priority brackets are nested inside lower-priority ones.

Input: String expression like `[{()}]` **Output:** true or false

Constraints:

- Priorities: `{` > `[` > `(`
- Length <= 10^4

Example: Input: `{[()]}` Output: true

Input: `[{()}]` Output: false

3. Parenthesis Depth Counter

Problem: Return the maximum depth of nested parentheses in a given string.

Input: String `s` **Output:** Integer depth

Constraints:

- s contains valid parentheses
- Length <= 10^4

Example: Input: `((a+(b))+(c+d))` Output: 3

4. Celebrity Problem with Liar

Problem: Among `n` people, one is a celebrity, and one person lies about whom they know. Find the celebrity.

Input: n x n matrix `knows[i][j]` **Output:** Index of celebrity or -1

Constraints:

- 2 <= n <= 100

Example: Input: knows = [[0,1,1],[0,0,1],[0,0,0]] Output: 2

5. Simulate Asteroid Collision

Problem: Given an array representing moving asteroids, return their final state after collisions.

Input: int[] asteroids **Output:** int[] final state

Constraints:

- -1000 <= asteroids[i] <= 1000
- 1 <= len <= 10^4

Example: Input: [5, 10, -5] Output: [5, 10]

6. Expression Simplifier

Problem: Simplify arithmetic expressions by removing parentheses and flattening signs.

Input: String `s` like `a-(b+c-(d+e))-f` **Output:** Flattened string

Constraints:

- Only lowercase letters and +/–/()`
- Length <= 10^4

Example: Input: `a-(b+c-(d+e))-f` Output: `a-b-c+d+e-f`

7. Remove Duplicate Letters Lexicographically Smallest

Problem: Remove duplicate letters to get smallest lexicographical order.

Input: String `s` **Output:** Modified string

Constraints:

- Only lowercase letters
- Length <= 10^4

Example: Input: `bcabc` Output: `abc`

8. Sliding Window Maximum Using Stack

Problem: Find the max of every window of size `k` using monotonic stack logic.

Input: int[] nums, int k **Output:** int[] of maximums

Constraints:

- 1 <= nums.length <= 10^5
- k <= nums.length

Example: Input: nums = [1,3,-1,-3,5,3,6,7], k = 3 Output: [3,3,5,5,6,7]

9. Spiral Order Matrix with Obstacles

Problem: Traverse a matrix in spiral order, skipping obstacles (marked as `-1`).

Input: int[][] matrix **Output:** int[] spiral order traversal

Constraints:

- 1 <= m,n <= 100

Example: Input: [[1,2,3], [4,-1,6], [7,8,9]] Output: [1,2,3,6,9,8,7,4]

10. Find Original Array from Doubled Array

Problem: Given an array that was created by doubling elements of an original array and shuffling, recover the original.

Input: int[] changed **Output:** int[] original

Constraints:

- 1 <= changed.length <= 10^5

Example: Input: [1,3,4,2,6,8] Output: [1,3,4]

11. Partition Array into Equal Halves with One Swap

Problem: Determine if two halves of an even-length array can be made equal by one swap.

Input: int[] arr **Output:** true or false

Constraints:

- Length % 2 == 0
- 2 <= length <= 10^5

Example: Input: [1,2,3,4] Output: true (swap 2 and 3)

12. Product of Array Except Self (No Extra Space)

Problem: Return product of all elements except self, without using division or extra space.

Input: int[] nums **Output:** int[] product

Constraints:

- 2 <= nums.length <= 10^5

Example: Input: [1,2,3,4] Output: [24,12,8,6]

13. Set Matrix Zeroes In-Place (No Flags)

Problem: Set entire row and column to 0 if element is 0, but cannot use first row/column as marker.

Input: int[][] matrix **Output:** Modified matrix

Constraints:

- 1 <= m,n <= 200

Example: Input: [[1,2,3], [4,0,6], [7,8,9]] Output: [[1,0,3], [0,0,0], [7,0,9]]

14. Sort Array by Parity and Index

Problem: Place even numbers at even indices and odd numbers at odd indices.

Input: int[] arr **Output:** Modified arr

Constraints:

- 1 <= arr.length <= 10^5

Example: Input: [3,1,2,4] Output: [2,1,4,3]

15. Shortest Unique Substring Containing All Characters

Problem: Return the smallest substring containing all unique characters in string.

Input: String `s` **Output:** Shortest substring

Constraints:

- Only lowercase letters
- Length <= 10^4

Example: Input: `aabcbcdbca` Output: `dbca`

16. Rearrange String to Avoid Adjacent Duplicates

Problem: Rearrange to avoid adjacent identical characters.

Input: String s **Output:** Rearranged string or ""

Constraints:

- Only lowercase letters
- Length <= 10^5

Example: Input: `aaabc` Output: `abaca`

17. Find and Replace Pattern Matching

Problem: Return all words matching a given pattern of letter positions.

Input: String[] words, String pattern **Output:** List of words

Constraints:

- Length of pattern <= 100
- 1 <= words.length <= 1000

Example: Input: words = ["mee", "aqq", "dkd", "ccc"], pattern = "abb" Output: ["mee", "aqq"]

18. Minimum Insertions to Make String Palindrome

Problem: Find the minimum number of insertions required to make a string palindrome.

Input: String s **Output:** int (min insertions)

Constraints:

- Only lowercase letters
- Length <= 500

Example: Input: `abcaa` Output: 1

19. Multiply Strings Without Integer Conversion

Problem: Multiply two large numbers represented as strings.

Input: String num1, String num2 **Output:** Product as string

Constraints:

- No built-in bigint or integer parsing
- Length <= 500

Example: Input: `"123"`, `"456"` Output: `“56088"`

20. Longest Substring with Each Char at Least K Times

Problem: Return the length of longest substring where each character appears at least `k` times.

Input: String s, int k **Output:** int length

Constraints:

- 1 <= s.length <= 10^4
- 1 <= k <= 100

Example: Input: s = "aaabb", k = 3 Output: 3